



## AQ9.5: Activity Questions 5 - Not Graded



This assignment will not be graded and is only for practice.

### Level 1

1 point

Let  $f(x, y, z) = (xy, x + y + z^2)$ , where  $x, y, z \in \mathbb{R}$ . Choose the set of correct options.

☐

The domain of  $f$  is  $\mathbb{R}^2$  and the co-domain of  $f$  is  $\mathbb{R}^2$

☐

The domain of  $f$  is  $\mathbb{R}^3$  and the co-domain of  $f$  is  $\mathbb{R}^2$

☐

$f$  is continuous everywhere in its domain

☐

$f$  is not continuous anywhere in its domain

**No, the answer is incorrect.**

**Score: 0**

### Accepted Answers:

The domain of  $f$  is  $\mathbb{R}^3$  and the co-domain of  $f$  is  $\mathbb{R}^2$

$f$  is continuous everywhere in its domain

1 point

Let  $f(x, y) = \begin{cases} \frac{\cos y \sin x}{x} & x \neq 0 \\ \cos y & x = 0 \end{cases}$ ,  $f(x, y) = \begin{cases} x \cos y \sin x & x \neq 0 \\ 0 & x = 0 \end{cases}$ .

Choose the correct options.

☐

$f$  is not defined at  $(0, 0)$

☐

$f$  is defined at  $(0, 0)$

☐

$f$  is continuous at  $(0, 0)$

☐

$f$  is not continuous at  $(0, 0)$

**No, the answer is incorrect.**

**Score: 0**

### Accepted Answers:

$f$  is defined at  $(0, 0)$

$f$  is continuous at  $(0, 0)$

1 point

Let  $h(u, v) = \sin(uv)$ ,  $g(x) = x^3$ ,  $f(y) = \cos y$ ,  $h(u, v) = \sin(uv)$ ,  $g(x) = x^3$ ,  $f(y) = \cos y$ . Let  $p(x, y) = h(g(x), f(y))$

Choose the correct option(s).

☐

$p$  is a scalar-valued function

☐

$p$  is a vector-valued function

☐


$pp$  is continuous everywhere since  $g.f$   $g.f$  is continuous everywhere in its domain

☐

$pp$  is continuous everywhere,  $gg$  is also continuous everywhere while  $ff$  is not continuous everywhere in its domain

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

$pp$  is a scalar-valued function

$pp$  is continuous everywhere since  $g.f$   $g.f$  is continuous everywhere in its domain

1 point

Let  $p(t) = (\ln(1 - t), \frac{1}{t}, 3t)$   $p(t) = (\ln(1 - t), t1, 3t)$ .

Choose the incorrect option(s).

☐

$pp$  is continuous for all real  $t$  less than 11 but not equal to zero

☐

$pp$  is continuous on  $\mathbb{R}$

☐

$pp$  is continuous in  $\{t \in \mathbb{R} | t < 1\}$   $\{t \in \mathbb{R} | t < 1\}$

☐

$pp$  is continuous in  $\{t \in \mathbb{R} | t \neq 1, t \neq 0\}$   $\{t \in \mathbb{R} | t = 1, t = 0\}$

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

$pp$  is continuous on  $\mathbb{R}$

$pp$  is continuous in  $\{t \in \mathbb{R} | t < 1\}$   $\{t \in \mathbb{R} | t < 1\}$

$pp$  is continuous in  $\{t \in \mathbb{R} | t \neq 1, t \neq 0\}$   $\{t \in \mathbb{R} | t = 1, t = 0\}$

1 point

Let  $f(x, y) = \ln(x^2 + y^2 - 1)$   $f(x, y) = \ln(x^2 + y^2 - 1)$ . Identify where  $ff$  is continuous. Note:

Equation of a circle with radius  $r$  and center  $(a, b)$  is given by  $(x - a)^2 + (y - b)^2 = r^2$

$(x - a)^2 + (y - b)^2 = r^2$

☐

Everywhere in  $\mathbb{R}^2$

☐

Everywhere in  $\mathbb{R}^2 \setminus \{(1, 0), (0, 1)\}$   $\mathbb{R}^2 \setminus \{(1, 0), (0, 1)\}$

☐

Everywhere in  $\mathbb{R}^2$  outside the circle with radius 1 and center  $(0, 0)$

☐ None of the above

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

Everywhere in  $\mathbb{R}^2$  outside the circle with radius 1 and center  $(0, 0)$

**Level 2**

1 point

Let  $g(u, v, w) = \frac{uv}{w}$   $g(u, v, w) = wuv$  and  $f(x, y, z) = g(P(x, y, z), Q(x, y, z), R(x, y, z))$

$f(x, y, z) = g(P(x, y, z), Q(x, y, z), R(x, y, z))$

where  $P(x, y, z) = e^{x^2 + y}$ ,  $Q(x, y, z) = \sqrt{y^2 + z^2 + 3}$ ,  $R(x, y, z) = \sin(xyz) + 5$

$$P(x, y, z) = ex^{2+y}, Q(x, y, z) = y^2 + z^2 + 3$$

$$\sqrt{\phantom{x}}, R(x, y, z) = \sin(xyz) + 5.$$

Choose the correct option(s).

☐

$f$  is a vector-valued function

☐

$f$  is a scalar-valued function

☐

$f$  is continuous everywhere in its domain.

☐

$P$  is continuous everywhere in its domain, while  $Q$  is not continuous everywhere in its domain

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

$f$  is a scalar-valued function

$f$  is continuous everywhere in its domain.

1 point

Let

$$f: \mathbb{R}^2 \rightarrow \mathbb{R}, f(x, y) = \begin{cases} \frac{x^2}{x^2 + y^2} & (x, y) \neq (0, 0) \\ 0 & (x, y) = (0, 0) \end{cases} \quad f: \mathbb{R}^2 \rightarrow \mathbb{R}, f(x, y) = \begin{cases} x^2 + y^2 & (x, y) \neq (0, 0) \\ 0 & (x, y) = (0, 0) \end{cases}$$

Choose the correct option(s).

☐

$\lim_{(x, y) \rightarrow (0, 0)} f$  does not exist

☐

$\lim_{(x, y) \rightarrow (0, 0)} f$  exists

☐

$f$  is continuous at  $(0, 0)$

☐

$f$  is not continuous at  $(0, 0)$

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

$\lim_{(x, y) \rightarrow (0, 0)} f$  does not exist

$f$  is not continuous at  $(0, 0)$

1 point

Let

$$f: \mathbb{R}^2 \rightarrow \mathbb{R}, f(x, y) = \begin{cases} \frac{xy}{x^2 + y^2} & (x, y) \neq (0, 0) \\ 0 & (x, y) = (0, 0) \end{cases} \quad f: \mathbb{R}^2 \rightarrow \mathbb{R}, f(x, y) = \begin{cases} x^2 + y^2 & (x, y) \neq (0, 0) \\ 0 & (x, y) = (0, 0) \end{cases}$$

Choose the correct option(s).

☐

$\lim_{(x, y) \rightarrow (0, 0)} f$  does not exist

☐

$\lim_{(x,y) \rightarrow (0,0)} f$  exists

☐

$f$  is continuous at  $(0,0)$

☐

$f$  is not continuous at  $(0,0)$

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

$\lim_{(x,y) \rightarrow (0,0)} f$  does not exist

$f$  is not continuous at  $(0,0)$

1 point

Let  $f(r) = \frac{1}{\langle r, r \rangle - 1}$ , where  $r = (x, y, z)$  and the inner product  $\langle, \rangle$  is the dot product  
Choose the set of incorrect options.

☐

$f$  is continuous throughout  $\mathbb{R}^3$

☐

$f$  is continuous throughout  $\mathbb{R}^3 \setminus \{(0, 0, 0)\}$

☐

$f$  is discontinuous on the unit sphere  $\{r \in \mathbb{R}^3 : \|r\| = 1\}$

☐

$f$  is continuous throughout  $\mathbb{R}^3 \setminus \{(1, 0, 0)\}$

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

$f$  is continuous throughout  $\mathbb{R}^3$

$f$  is continuous throughout  $\mathbb{R}^3 \setminus \{(0, 0, 0)\}$

$f$  is continuous throughout  $\mathbb{R}^3 \setminus \{(1, 0, 0)\}$

Let  $f: \mathbb{R}^2 \rightarrow \mathbb{R}$ ,  $f(x, y) = \begin{cases} \frac{x^k y}{x^4 + y^4} & (x, y) \neq (0, 0) \\ 0 & (x, y) = (0, 0) \end{cases}$

$(0, 0)$ . Find the minimum possible positive integer value of  $k$  such that  $f$  is continuous at  $(0, 0)$ .

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

(Type: Numeric) 4

1 point

[Check Answers](#)

Your score is: 0/10